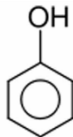
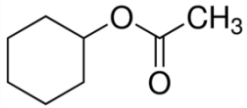


Section A

Question				Marking details	Marks available											
					AO1	AO2	AO3	Total	Maths	Prac						
1.				<table><tr><td>Reagent</td><td>Observation</td></tr><tr><td>iron(III) chloride</td><td>purple coloration / solution</td></tr><tr><td>sodium hydroxide</td><td>no observation / change</td></tr></table>	Reagent	Observation	iron(III) chloride	purple coloration / solution	sodium hydroxide	no observation / change	2			2		2
Reagent	Observation															
iron(III) chloride	purple coloration / solution															
sodium hydroxide	no observation / change															
2.				C ₆ H ₈ O ₃		1		1								
3.				 + Na₂CO₃ accept C₆H₅ONa + NaHCO₃		1		1								
4.				sodium tetrahydridoborate(III) / sodium borohydride / NaBH ₄	1			1		1						
5.				total relative peak area = 36 + 18 + 13 = 67 % 1,3-dimethylbenzene = 18 × 100 / 67 = 26.87 = 27		1		1	1							
6.				1-propyl ethanoate (accept propyl ethanoate)		1		1								

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7.						1		1		
8.				A nitric(III) acid / HNO_2 / sodium nitrate(III) and hydrochloric acid / NaNO_2 and HCl (1) B acidified dichromate / $\text{Cr}_2\text{O}_7^{2-}$ and H^+ (1)	2					2
				Section A total	5	5	0	10	1	5